

# CampusConnect

## *A Smart Campus Exchange Platform*

### Overview

This document presents the project execution plan for CampusConnect, covering the team-level timeline and milestones, module ownership and task assignments, individual execution plans and checkpoints, cross-module dependencies, and the tools used to plan and track work. It builds on the scope, technology stack, and roles defined in the Project Proposal, and translates them into a week-by-week execution plan across the 14-week development cycle. This revision updates the team-level phase timeline and the Gantt chart to reflect the current schedule, and adds a granular, subtask-level breakdown of every task on the board alongside its tracking status.

### Team Task Timeline & Milestones

The team-level timeline is organized into nine phases, each with a defined duration and set of deliverables. Phases overlap where backend, frontend, AI integration, and testing work can proceed in parallel, allowing module leads to begin dependent tasks as soon as the required APIs or designs are available. Module-level testing now runs continuously alongside development, with a dedicated final integration, security, and regression pass ahead of deployment.

| Phase   | Milestone                                        | Duration     | Key Deliverables                                                                                                                                    |
|---------|--------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 | Planning & Requirement Analysis                  | Week 1 - 3   | Finalized scope, requirements gathering, wireframing, and technology setup, including AI/OCR feasibility research.                                  |
| Phase 2 | UI/UX Design                                     | Week 2 - 4   | Design system, wireframes to high-fidelity mockups, and design review.                                                                              |
| Phase 3 | Core Backend Development                         | Week 3 - 10  | Database schema, authentication, listing CRUD APIs, admin APIs, and ongoing backend integration support through Week 10.                            |
| Phase 4 | Core Frontend Development                        | Week 5 - 9   | React components for browsing, listing creation, chat UI, and dashboards.                                                                           |
| Phase 5 | AI Feature Integration                           | Week 6 - 11  | AI-assisted listing creation, intelligent matching engine, and OCR duplicate detection.                                                             |
| Phase 6 | Real-Time Features & Notifications               | Week 9 - 12  | In-app chat, real-time notifications, and rental reminders via Socket.io.                                                                           |
| Phase 7 | Continuous Module-Level Testing                  | Week 5 - 13  | Ongoing unit and module-level testing performed by each lead in parallel with development, enabling early bug detection ahead of final sign-off.    |
| Phase 8 | Final Integration, Security & Regression Testing | Week 11 - 13 | Cross-module integration testing, security checks, and regression testing across the full platform ahead of deployment.                             |
| Phase 9 | Deployment & Final Presentation                  | Week 9 - 14  | Performance monitoring and deployment-pipeline preparation from Week 9, culminating in cloud deployment, documentation, and the final project demo. |

The following milestones mark the completion of each major capability and are used to track whether the team remains on schedule:

- Requirements, architecture, and technology stack finalized — end of Week 2.
- High-fidelity UI/UX design approved — end of Week 4.
- Core backend APIs (authentication, listings, admin) live — end of Week 7.
- Core frontend integrated with backend services — end of Week 9.
- AI-assisted listing creation and matching engine integrated — end of Week 10.
- Real-time chat and notification system live — end of Week 11.
- Testing and QA sign-off across all modules — end of Week 13.
- Final deployment and project demonstration — Week 14.
- AI/OCR feasibility research and requirement gathering continue through Week 3 as part of Planning & Requirement Analysis.
- Continuous module-level testing runs in parallel with development from Week 5 through Week 13.
- Deployment-pipeline preparation and performance monitoring begin as early as Week 9, ahead of final release.

## Team Task Assignments

Each module has one accountable owner. In addition, every member is a named deputy for one other module that depends closely on their own work, so no module relies on a single point of contact if its owner is temporarily blocked or unavailable.

| Team Member    | Module Owned                                                                                                                                                                                                          | Deputy / Support For                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Dev Barot      | <b>Project Manager &amp; System Architect</b> — Project planning, system architecture, module integration, task coordination, overall execution, audit logging, error handling, rate limiting, and system monitoring. | <b>Overall project integration</b> — Supports architecture, integration, deployment, and cross-module decisions. |
| Dhrumil Doshi  | <b>Authentication &amp; User Management</b> — Secure authentication, user profiles, role-based access, ratings & reputation, payment integration, transaction receipts, and sustainability dashboard.                 | <b>Kajal Varlani</b> — Admin access control, user management, and role checks.                                   |
| Nishka Shah    | <b>Marketplace &amp; Listing Management</b> — Item categories, listing CRUD, image uploads, listing UI, category management, AI-assisted listing integration, and testing of marketplace workflows.                   | <b>Charmi Bhayani</b> — Listing-to-rental and transaction workflow continuity.                                   |
| Heer Mehta     | <b>AI &amp; Intelligent Services</b> — AI/OCR research, FastAPI AI service, AI listing assistance, price estimation, semantic matching, OCR duplicate detection, Redis Pub/Sub integration, and matching accuracy.    | <b>Nishka Shah</b> — AI-assisted listing creation, suggestions, and marketplace integration.                     |
| Heer Shah      | <b>Search, Chat &amp; Notification</b> — Search and filtering APIs, advanced search, saved searches, wishlist, match/rental notifications, Socket.io chat, and real-time notifications.                               | <b>Heer Mehta</b> — Match-event triggers and AI matching notification integration.                               |
| Charmi Bhayani | <b>Rental &amp; Transaction Management</b> — Rental APIs, transaction management, QR-code handover verification, rental dashboard, due-date tracking, and rental reminders.                                           | <b>Heer Shah</b> — Rental reminder notifications and notification pipeline integration.                          |
| Kajal Varlani  | <b>Admin Panel &amp; Analytics</b> — Admin APIs, listing approval, user management, dispute handling, admin dashboard, analytics, reporting, moderation notifications, and testing.                                   | <b>Deep Boghara</b> — Analytics data pipeline, reporting data, and testing support.                              |
| Deep Boghara   | <b>Database, DevOps &amp; Quality Assurance</b> — Database design, MongoDB Atlas, Git/CI pipeline, automated testing, integration support, performance and security testing, and deployment.                          | <b>Dev Barot</b> — Architecture, infrastructure, integration, and deployment decisions.                          |

Individual Team Member Task Assignments

Dev Barot (202512069) — Project Manager & System Architect

| Phase    | Weeks | Task                                                                                                                                        |
|----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Planning | W1-2  | Finalize project requirements and define the overall project scope.                                                                         |
| Planning | W1-2  | Define the system architecture and overall technical structure of the project.                                                              |
| Planning | W1-2  | Set up the Git repository and establish the project's version-control workflow.                                                             |
| Planning | W1-2  | Set up the project task board for task assignment, tracking, and coordination.                                                              |
| Planning | W2    | Define cross-module API contract standards and prepare the OpenAPI skeleton to provide shared contracts for independent module development. |
| Design   | W3-4  | Review UI/UX design deliverables and ensure that designs align with the approved project requirements.                                      |
| Design   | W3-4  | Validate the system architecture against the approved wireframes and UI/UX structure.                                                       |
| Design   | W3-4  | Implement environment and secrets configuration management using <code>.env</code> / configuration services.                                |
| Backend  | W4-5  | Set up a centralized logging framework across services using Winston/Pino.                                                                  |
| Backend  | W5-6  | Build audit-logging middleware to capture important user and administrator actions across the system.                                       |
| Backend  | W5-6  | Build the audit-log storage schema along with query and filtering APIs for retrieving audit records.                                        |

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| <b>Backend</b>   | <b>W6</b>     | Implement global error-handling and exception-tracking middleware for consistent error management across the system.                       |
| <b>Backend</b>   | <b>W6-7</b>   | Implement API rate-limiting and request-throttling middleware to control excessive API requests and improve system security and stability. |
| <b>Testing</b>   | <b>W7</b>     | Test the audit-logging and global error-handling middleware as an early module-level checkpoint.                                           |
| <b>Backend</b>   | <b>W5-9</b>   | Coordinate backend and frontend integration across all project modules and resolve integration dependencies.                               |
| <b>Backend</b>   | <b>W5-9</b>   | Conduct sprint reviews with module leads to track progress, identify issues, and coordinate development activities.                        |
| <b>Backend</b>   | <b>W7</b>     | Conduct a mid-project architecture review with all module leads to verify architectural consistency and integration.                       |
| <b>Backend</b>   | <b>W7</b>     | Update and maintain the project risk log based on identified technical, integration, and execution risks.                                  |
| <b>AI</b>        | <b>W8-10</b>  | Oversee the integration of AI services into the core system architecture and ensure compatibility with existing modules.                   |
| <b>AI</b>        | <b>W9-10</b>  | Implement health-check and uptime-monitoring endpoints for all major services to monitor system availability and health.                   |
| <b>Real-Time</b> | <b>W10-11</b> | Coordinate the rollout and integration of real-time features across all modules.                                                           |
| <b>Testing</b>   | <b>W10-11</b> | Test API contract compliance and health-check endpoints across modules as a mid-project integration checkpoint.                            |
| <b>Testing</b>   | <b>W11-13</b> | Lead overall testing coordination across all module teams and ensure testing activities are completed as planned.                          |
| <b>Testing</b>   | <b>W11-13</b> | Track bug identification, assignment, resolution, and verification across teams.                                                           |

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| <b>Testing</b>    | <b>W13</b> | Run a full-system regression test before deployment to verify that existing functionality remains stable after integration.                    |
| <b>Deployment</b> | <b>W14</b> | Oversee the final system deployment and ensure that all modules and services are properly deployed and integrated.                             |
| <b>Deployment</b> | <b>W14</b> | Prepare the final project documentation and presentation covering the completed system, architecture, implementation, testing, and deployment. |

**Dhrumil Doshi (202512088) — Authentication & User Management Lead**

| Phase           | Weeks       | Task                                                                                            |
|-----------------|-------------|-------------------------------------------------------------------------------------------------|
| <b>Planning</b> | <b>W1–2</b> | Gather requirements for authentication, user roles, and access levels.                          |
| <b>Planning</b> | <b>W2</b>   | Gather requirements for payment methods and the personal sustainability dashboard.              |
| <b>Design</b>   | <b>W2–3</b> | Design login, signup, and profile UI wireframes.                                                |
| <b>Design</b>   | <b>W3–4</b> | Design payment method and receipt-confirmation wireframes.                                      |
| <b>Design</b>   | <b>W3–4</b> | Design personal sustainability dashboard wireframe.                                             |
| <b>Backend</b>  | <b>W3–5</b> | Develop JWT-based authentication, password handling, and role-based access control (RBAC) APIs. |
| <b>Frontend</b> | <b>W5–6</b> | Build user profile and account settings frontend components.                                    |
| <b>Testing</b>  | <b>W5</b>   | Test authentication and RBAC APIs at the module checkpoint.                                     |
| <b>Backend</b>  | <b>W5–6</b> | Build payment-method storage and payment-gateway integration APIs.                              |

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| <b>Backend</b>    | <b>W5–6</b>   | Build item-receipt confirmation API with separate buyer and seller confirmations.                                             |
| <b>Backend</b>    | <b>W6–7</b>   | Build the personal sustainability dashboard backend for money saved, items reused, and environmental/CO <sub>2</sub> impact.  |
| <b>Frontend</b>   | <b>W6–7</b>   | Build payment method management UI.                                                                                           |
| <b>Frontend</b>   | <b>W7–8</b>   | Build personal sustainability dashboard UI.                                                                                   |
| <b>Testing</b>    | <b>W8</b>     | Test profile, payment, and sustainability dashboard UI at the module checkpoint.                                              |
| <b>Backend</b>    | <b>W7–9</b>   | Support marketplace and rental modules with authentication checks and access-control integration as deputy for Kajal Varlani. |
| <b>Backend</b>    | <b>W8–9</b>   | Integrate payment confirmation with rental and marketplace transaction flows.                                                 |
| <b>Real-Time</b>  | <b>W9–10</b>  | Integrate the ratings and reputation system with notification triggers.                                                       |
| <b>Real-Time</b>  | <b>W9–10</b>  | Trigger semester-end reminders encouraging students to list unused items.                                                     |
| <b>Testing</b>    | <b>W11–12</b> | Test authentication flows, payment and receipt-confirmation flows, and resolve security issues.                               |
| <b>Deployment</b> | <b>W14</b>    | Support deployment of authentication, payment, and sustainability services.                                                   |

## Nishka Shah (202512035) — Marketplace & Listing Management Lead

| Phase    | Weeks | Task                                                                                                                                           |
|----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Planning | W1-2  | Define item categories for the marketplace.                                                                                                    |
| Planning | W1-2  | Define listing workflow requirements for <b>buy, sell, rent, exchange, and giveaway</b> processes.                                             |
| Design   | W3-4  | Design the listing creation wireframe for creating and managing marketplace listings.                                                          |
| Design   | W3-4  | Design the browsing wireframe for discovering and viewing marketplace listings.                                                                |
| Backend  | W4-6  | Develop listing CRUD APIs for creating, reading, updating, and deleting marketplace listings.                                                  |
| Backend  | W4-6  | Develop category management functionality for organizing and managing marketplace item categories.                                             |
| Backend  | W6-7  | Develop the image-upload backend to support images associated with marketplace listings.                                                       |
| Testing  | W7    | Test listing CRUD and image-upload APIs as an early module-level checkpoint.                                                                   |
| Frontend | W6-8  | Build the listing creation UI and integrate it with the listing APIs.                                                                          |
| Frontend | W6-8  | Build the browsing UI for viewing and exploring marketplace listings.                                                                          |
| Frontend | W8-9  | Build category UI components for displaying and navigating marketplace categories.                                                             |
| Backend  | W8-9  | Coordinate with Charmi Bhayani to ensure listings transition cleanly into rental workflows and maintain listing-to-rental workflow continuity. |

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| <b>AI</b>         | <b>W9-11</b>  | Integrate AI-assisted listing suggestions into the listing creation flow, working with Heer Mehta and using mocked AI responses initially so development is not blocked by the AI service. |
| <b>Testing</b>    | <b>W10-11</b> | Test the listing creation and browsing UI end-to-end as an early module-level checkpoint.                                                                                                  |
| <b>Testing</b>    | <b>W11-13</b> | Test the <b>buy workflow</b> end-to-end to verify the complete marketplace process.                                                                                                        |
| <b>Testing</b>    | <b>W11-13</b> | Test the <b>sell workflow</b> end-to-end to verify listing and selling functionality.                                                                                                      |
| <b>Testing</b>    | <b>W11-13</b> | Test the <b>rent workflow</b> end-to-end, including the transition from marketplace listing to rental functionality.                                                                       |
| <b>Testing</b>    | <b>W11-13</b> | Test the <b>exchange workflow</b> end-to-end.                                                                                                                                              |
| <b>Testing</b>    | <b>W11-13</b> | Test the <b>giveaway workflow</b> end-to-end.                                                                                                                                              |
| <b>Deployment</b> | <b>W14</b>    | Finalize the marketplace and listing management module for deployment and ensure its integration with the complete system.                                                                 |

### Heer Mehta (202512095) — AI & Intelligent Services Lead

| <b>Phase</b>    | <b>Weeks</b> | <b>Task</b>                                                                                      |
|-----------------|--------------|--------------------------------------------------------------------------------------------------|
| <b>Planning</b> | <b>W1-3</b>  | Research AI requirements and identify the AI capabilities required for the CampusConnect system. |
| <b>Planning</b> | <b>W1-3</b>  | Research OCR requirements for duplicate detection of notes and question papers.                  |



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| <b>Planning</b>  | <b>W1-3</b>   | Evaluate the feasibility of using the Groq API for the project's AI requirements.                                                                        |
| <b>Backend</b>   | <b>W4-6</b>   | Set up the FastAPI AI service scaffolding using mocked/sample listing data so AI service development is not blocked by the completion of backend models. |
| <b>Testing</b>   | <b>W6</b>     | Test the FastAPI AI service scaffolding using mock requests as an early module-level checkpoint.                                                         |
| <b>AI</b>        | <b>W6-10</b>  | Develop AI-assisted listing creation to help users generate and improve marketplace listings using AI.                                                   |
| <b>AI</b>        | <b>W6-10</b>  | Develop the AI-based price estimation functionality for marketplace listings.                                                                            |
| <b>AI</b>        | <b>W6-10</b>  | Develop the semantic matching engine to identify relevant matches between listings and user requirements.                                                |
| <b>AI</b>        | <b>W6-10</b>  | Develop OCR-based duplicate detection for notes and question papers.                                                                                     |
| <b>Testing</b>   | <b>W9</b>     | Test OCR duplicate-detection accuracy using sample notes and question-paper documents as an early module-level checkpoint.                               |
| <b>AI</b>        | <b>W9-10</b>  | Support Nishka Shah in refining and integrating AI-assisted listing suggestions into the listing creation workflow as part of the deputy pairing.        |
| <b>Real-Time</b> | <b>W10-11</b> | Integrate match-event triggers with the Redis Pub/Sub notification pipeline so that relevant matching events can be consumed by the notification system. |
| <b>Testing</b>   | <b>W10-11</b> | Prepare an evaluation dataset for measuring the precision of the AI semantic matching engine.                                                            |
| <b>Testing</b>   | <b>W10-11</b> | Prepare a benchmark report documenting the AI matching precision and evaluation results.                                                                 |
| <b>Testing</b>   | <b>W11-13</b> | Test AI recommendation accuracy to verify the relevance and quality of AI-generated recommendations.                                                     |

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| <b>Testing</b>    | <b>W11-13</b> | Test matching precision using the prepared evaluation dataset and benchmark the performance of the semantic matching engine.                                              |
| <b>Deployment</b> | <b>W14</b>    | Finalize the AI service for deployment, including integration and deployment readiness of the AI-assisted listing, price estimation, semantic matching, and OCR services. |

### Heer Shah (202512024) — Search, Chat & Notification Lead

| <b>Phase</b>    | <b>Weeks</b> | <b>Task</b>                                                                                                                                    |
|-----------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Planning</b> | <b>W1-2</b>  | Define search requirements, including the required search functionality for marketplace listings.                                              |
| <b>Planning</b> | <b>W1-2</b>  | Define filtering requirements for browsing and narrowing marketplace listings.                                                                 |
| <b>Planning</b> | <b>W1-2</b>  | Define notification requirements, including match events, rental reminders, real-time notifications, and notification delivery.                |
| <b>Design</b>   | <b>W3-4</b>  | Design the chat UI wireframe for the in-app messaging experience.                                                                              |
| <b>Design</b>   | <b>W3-4</b>  | Design the notification UI wireframe for displaying real-time and system notifications.                                                        |
| <b>Backend</b>  | <b>W4-6</b>  | Build search endpoints against seeded/sample listing data so search development is not blocked by the Marketplace & Listing Management module. |
| <b>Backend</b>  | <b>W4-6</b>  | Build filtering endpoints against seeded/sample listing data to support listing discovery and filtering independently of the listing module.   |
| <b>Testing</b>  | <b>W6-7</b>  | Test search and filtering endpoints as an early module-level checkpoint to verify API functionality and results.                               |
| <b>Frontend</b> | <b>W7-9</b>  | Build the advanced search UI and integrate it with the search endpoints.                                                                       |

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| <b>Frontend</b>   | <b>W7-9</b>   | Build the filters UI for refining marketplace search results.                                                                                                     |
| <b>Frontend</b>   | <b>W7-9</b>   | Build the saved-search UI to allow users to save and manage frequently used searches.                                                                             |
| <b>Frontend</b>   | <b>W7-9</b>   | Build the wishlist UI for users to save and manage desired listings.                                                                                              |
| <b>Testing</b>    | <b>W9</b>     | Test the search, filters, saved-search, and wishlist UI as a module-level checkpoint.                                                                             |
| <b>Real-Time</b>  | <b>W10-11</b> | Integrate match-event triggers from Heer Mehta's AI & Intelligent Services module into the shared notification pipeline using the defined event contract.         |
| <b>Real-Time</b>  | <b>W10-11</b> | Integrate rental due-date triggers from Charmi Bhayani's Rental & Transaction Management module into the notification pipeline for rental reminder notifications. |
| <b>Real-Time</b>  | <b>W10-12</b> | Implement Socket.io-based in-app chat to provide real-time communication between users.                                                                           |
| <b>Real-Time</b>  | <b>W10-12</b> | Implement real-time notifications and integrate them with the shared notification event pipeline.                                                                 |
| <b>Testing</b>    | <b>W12-13</b> | Test chat reliability, including real-time message delivery and communication functionality.                                                                      |
| <b>Testing</b>    | <b>W12-13</b> | Test notification delivery to ensure match, rental-reminder, and real-time notifications are triggered and delivered correctly.                                   |
| <b>Deployment</b> | <b>W14</b>    | Finalize the search, chat, and notification services and prepare them for deployment and integration with the complete system.                                    |

## Charmi Bhayani (202512028) — Rental & Transaction Management Lead

| Phase    | Weeks | Task                                                                                                                         |
|----------|-------|------------------------------------------------------------------------------------------------------------------------------|
| Planning | W1-2  | Define rental requirements, including rental requests, rental processes, due dates, reminders, and overall rental workflow.  |
| Planning | W1-2  | Define transaction management requirements, including transaction states and the flow of transactions throughout the system. |
| Design   | W3-4  | Design the rental request wireframe for submitting, viewing, and managing rental requests.                                   |
| Design   | W3-4  | Design the QR handover wireframe for secure item handover and verification.                                                  |
| Backend  | W5-7  | Develop rental request APIs for creating, managing, and processing rental requests.                                          |
| Backend  | W5-7  | Develop transaction state management to track and manage the different states of rental and transaction processes.           |
| Testing  | W7    | Test rental request APIs and transaction states as an early module-level checkpoint.                                         |
| Backend  | W7-8  | Integrate the QR code generation module to support secure rental handovers.                                                  |
| Backend  | W7-8  | Integrate the QR code verification module to validate and secure rental handovers.                                           |
| Frontend | W7-9  | Build the rental dashboard UI for managing rental requests, active rentals, and related rental information.                  |
| Frontend | W7-9  | Build the due-date tracking UI to display and manage rental due dates and deadlines.                                         |

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| <b>Testing</b>    | <b>W9-10</b>  | Test the complete QR handover flow and rental dashboard UI as an early module-level checkpoint.                                                                                       |
| <b>Real-Time</b>  | <b>W10-11</b> | Implement rental reminder scheduling using <b>node-cron/BullMQ</b> and send rental reminder events to Heer Shah's shared notification pipeline through the defined event integration. |
| <b>Testing</b>    | <b>W12-13</b> | Test rental workflows end-to-end to verify the complete rental process, including rental requests, handovers, due dates, and reminders.                                               |
| <b>Testing</b>    | <b>W12-13</b> | Test transaction workflows end-to-end to verify transaction state management and successful completion of transaction processes.                                                      |
| <b>Deployment</b> | <b>W14</b>    | Finalize the rental and transaction management module for deployment and ensure its integration with the complete system.                                                             |

**Kajal Varlani (202512017) — Admin Panel & Analytics Lead**

| <b>Phase</b>    | <b>Weeks</b> | <b>Task</b>                                                                                                                                      |
|-----------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Planning</b> | <b>W1-2</b>  | Define admin requirements, including administrative functions, access, listing management, user management, disputes, reporting, and moderation. |
| <b>Planning</b> | <b>W1-2</b>  | Define moderation workflows for listing approval, content moderation, user-related actions, and dispute handling.                                |
| <b>Design</b>   | <b>W3-4</b>  | Design admin dashboard wireframes covering administration, listing approval, user management, dispute handling, and analytics.                   |

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| <b>Backend</b>    | <b>W5-6</b>   | Develop the admin API for listing approval using seeded sample listings so moderation APIs can be developed and tested independently. |
| <b>Backend</b>    | <b>W5-6</b>   | Develop the admin API for user management to support administrative user-related operations.                                          |
| <b>Backend</b>    | <b>W6-7</b>   | Develop the admin API for dispute handling and managing dispute-related workflows.                                                    |
| <b>Testing</b>    | <b>W7</b>     | Test admin listing approval, user-management, and dispute-handling APIs as an early module-level checkpoint.                          |
| <b>Frontend</b>   | <b>W7-9</b>   | Build the admin dashboard UI and integrate it with the administrative APIs.                                                           |
| <b>Frontend</b>   | <b>W7-9</b>   | Build analytics views to present relevant system and marketplace data through the admin dashboard.                                    |
| <b>Testing</b>    | <b>W9</b>     | Test the admin dashboard and analytics UI using sample data as an early module-level checkpoint.                                      |
| <b>Backend</b>    | <b>W9-10</b>  | Coordinate with Deep Boghara on the analytics data pipeline that feeds data into the admin dashboard.                                 |
| <b>Real-Time</b>  | <b>W10-11</b> | Integrate reporting tools to support administrative reporting and analytics requirements.                                             |
| <b>Real-Time</b>  | <b>W10-11</b> | Integrate moderation notifications to notify relevant users or administrators about moderation-related actions and events.            |
| <b>Testing</b>    | <b>W12-13</b> | Test complete admin workflows, including listing approval, user management, moderation, and related administrative operations.        |
| <b>Testing</b>    | <b>W12-13</b> | Test dispute-resolution processes end-to-end to verify that disputes can be handled correctly through the system.                     |
| <b>Deployment</b> | <b>W14</b>    | Finalize the admin panel and analytics module for deployment and ensure its integration with the complete system.                     |

**Deep Boghara (202512076) — Database, DevOps & Quality Assurance Lead**

| Phase      | Weeks | Task                                                                                                                                                          |
|------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planning   | W1-2  | Design the database schema for the project, including the structure required to support all major modules.                                                    |
| Planning   | W1-2  | Set up MongoDB Atlas and configure the project's database environment.                                                                                        |
| Planning   | W1-2  | Set up the Git workflow, including branching strategy and pull-request rules for collaborative development.                                                   |
| Design     | W3-4  | Review the database schema against the approved UI/UX flows and validate that the data structure supports the planned application workflows.                  |
| Backend    | W3-4  | Set up the Continuous Integration (CI) pipeline for automated development and integration checks.                                                             |
| Backend    | W3-4  | Set up automated build checks to verify that code changes build successfully before integration.                                                              |
| Testing    | W5    | Set up the test-automation framework and integrate it with the CI pipeline as an early module-level testing checkpoint.                                       |
| Backend    | W5-8  | Support backend and frontend integration on an ongoing basis and help resolve integration issues between modules.                                             |
| Backend    | W5-8  | Manage version control across the team on an ongoing basis, including maintaining the Git workflow and coordinating code integration.                         |
| Testing    | W8    | Run module-level unit tests as each module reaches feature completion, providing early testing checkpoints rather than holding testing until the final stage. |
| Deployment | W9-11 | Monitor the performance of AI services and identify performance or reliability issues during integration.                                                     |
| Deployment | W9-11 | Monitor the performance of real-time services to ensure reliable operation and identify potential performance issues.                                         |

|                   |               |                                                                                                                                                 |
|-------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deployment</b> | <b>W9-11</b>  | Prepare the deployment pipeline and ensure the required infrastructure is ready for application deployment.                                     |
| <b>Testing</b>    | <b>W10-11</b> | Run cross-module integration tests as individual modules are connected, providing an early integration testing checkpoint.                      |
| <b>Testing</b>    | <b>W12-13</b> | Lead final unit testing across the project and ensure individual components meet functional requirements.                                       |
| <b>Testing</b>    | <b>W12-13</b> | Lead integration testing to verify that all modules and services work correctly together.                                                       |
| <b>Testing</b>    | <b>W12-13</b> | Lead security checks to identify security issues and validate the system before deployment.                                                     |
| <b>Deployment</b> | <b>W13</b>    | Support Dev Barot with the final architecture and release-readiness review as part of the deputy pairing before deployment.                     |
| <b>Deployment</b> | <b>W14</b>    | Manage final deployment across Vercel/Netlify, Render/AWS, and MongoDB Atlas and ensure the application and database are successfully deployed. |



## Individual Milestones

| Team Member          | Key Milestone                                                                                             | Target Week    |
|----------------------|-----------------------------------------------------------------------------------------------------------|----------------|
| <b>Dev Barot</b>     | Architecture, Git repository, and project task board finalized                                            | <b>Week 2</b>  |
| Dev Barot            | Cross-module API contracts and development environment established                                        | <b>Week 4</b>  |
| Dev Barot            | Backend and frontend integration coordinated across modules                                               | <b>Week 9</b>  |
| Dev Barot            | AI and real-time service integration coordinated across the system                                        | <b>Week 11</b> |
| Dev Barot            | Full-system regression testing completed ahead of deployment                                              | <b>Week 13</b> |
| Dev Barot            | Final deployment and project documentation/presentation delivered                                         | <b>Week 14</b> |
| <b>Dhrumil Doshi</b> | Authentication, password handling, and RBAC APIs live                                                     | <b>Week 5</b>  |
| Dhrumil Doshi        | Payment and receipt-confirmation services integrated with transaction flows                               | <b>Week 9</b>  |
| Dhrumil Doshi        | Ratings and reputation system integrated with notification triggers                                       | <b>Week 10</b> |
| Dhrumil Doshi        | Authentication, payment, and receipt-confirmation flows fully tested and security issues resolved         | <b>Week 12</b> |
| <b>Nishka Shah</b>   | Listing CRUD, category management, and image-upload APIs completed                                        | <b>Week 7</b>  |
| Nishka Shah          | Listing creation, browsing, and category UI completed                                                     | <b>Week 9</b>  |
| Nishka Shah          | AI-assisted listing suggestions integrated                                                                | <b>Week 11</b> |
| Nishka Shah          | Buy, sell, rent, exchange, and giveaway workflows fully tested                                            | <b>Week 13</b> |
| <b>Heer Mehta</b>    | FastAPI AI service scaffolding ready and validated with mock requests                                     | <b>Week 6</b>  |
| Heer Mehta           | AI-assisted listing, price estimation, semantic matching, and OCR duplicate-detection features functional | <b>Week 10</b> |
| Heer Mehta           | Match-event triggers integrated with Redis pub/sub notification pipeline                                  | <b>Week 11</b> |
| Heer Mehta           | AI recommendation accuracy and matching precision validated                                               | <b>Week 13</b> |

|                       |                                                                                                         |                |
|-----------------------|---------------------------------------------------------------------------------------------------------|----------------|
| <b>Heer Shah</b>      | Search and filtering APIs completed and tested                                                          | <b>Week 7</b>  |
| Heer Shah             | Advanced search, filters, saved searches, and wishlist UI completed                                     | <b>Week 9</b>  |
| Heer Shah             | Match-event and rental reminder triggers integrated into notification pipeline                          | <b>Week 11</b> |
| Heer Shah             | In-app chat and real-time notification services live                                                    | <b>Week 12</b> |
| Heer Shah             | Chat reliability and notification delivery fully tested                                                 | <b>Week 13</b> |
| <b>Charmi Bhayani</b> | Rental request APIs and transaction state management completed                                          | <b>Week 7</b>  |
| Charmi Bhayani        | QR generation and verification handover module live                                                     | <b>Week 8</b>  |
| Charmi Bhayani        | Rental dashboard and due-date tracking UI completed                                                     | <b>Week 9</b>  |
| Charmi Bhayani        | Rental reminder scheduling live                                                                         | <b>Week 11</b> |
| Charmi Bhayani        | Rental and transaction workflows fully tested                                                           | <b>Week 13</b> |
| <b>Kajal Varlani</b>  | Admin approval, user-management, and dispute APIs completed and tested                                  | <b>Week 7</b>  |
| Kajal Varlani         | Admin dashboard and analytics views completed                                                           | <b>Week 9</b>  |
| Kajal Varlani         | Analytics data pipeline coordinated with Deep Boghara and reporting/moderation notifications integrated | <b>Week 11</b> |
| Kajal Varlani         | Admin workflows and dispute-resolution processes fully tested                                           | <b>Week 13</b> |
| <b>Deep Boghara</b>   | Database schema, MongoDB Atlas, and Git branching/PR workflow set up                                    | <b>Week 2</b>  |
| Deep Boghara          | CI pipeline and automated build checks ready                                                            | <b>Week 4</b>  |
| Deep Boghara          | Cross-module integration testing and deployment pipeline prepared                                       | <b>Week 11</b> |
| Deep Boghara          | Final deployment executed across all services                                                           | <b>Week 14</b> |

## Dependencies & Handoffs

The schedule above is shaped by a small number of key handoffs between members. These are the dates that matter most for staying on track, since a delay in any of them affects the modules that depend on it:

| From           | To                                         | Handoff                                                                                       | Week    |
|----------------|--------------------------------------------|-----------------------------------------------------------------------------------------------|---------|
| Deep Boghara   | All Members                                | Database schema, MongoDB environment, and Git/branching workflow ready for module development | Week 2  |
| Deep Boghara   | All Members                                | CI pipeline and automated build checks ready for shared integration                           | Week 4  |
| Dhrumil Doshi  | Nishka Shah, Charmi Bhayani, Kajal Varlani | Authentication APIs, password handling, and RBAC ready for module integration                 | Week 5  |
| Nishka Shah    | Heer Mehta                                 | Listing data/API structure available for AI-assisted listing and matching integration         | Week 7  |
| Nishka Shah    | Charmi Bhayani                             | Listings ready for clean transition into rental workflows                                     | Week 9  |
| Heer Mehta     | Nishka Shah                                | AI-assisted listing suggestions available for listing-creation integration                    | Week 10 |
| Heer Mehta     | Heer Shah                                  | Match-event triggers ready for the notification pipeline                                      | Week 11 |
| Charmi Bhayani | Heer Shah                                  | Rental due-date/reminder events ready for notification integration                            | Week 11 |
| Dhrumil Doshi  | Marketplace & Rental Modules               | Payment confirmation integrated into marketplace and rental transaction flows                 | Week 9  |
| Kajal Varlani  | Dev Barot                                  | Analytics and moderation data/reporting available for final system reporting                  | Week 12 |
| Deep Boghara   | Dev Barot                                  | Release-readiness, integration testing, and deployment infrastructure prepared                | Week 13 |

Timeline Chart & Task Board

The timeline chart and task board referenced throughout this document are maintained in GanttProject-style and Trello respectively. The current state of each is captured below; the live board and the exported chart are also submitted separately as described in Submission Deliverables.

Phase Overview

| Phase   | Milestone                                        | Week Range   | Color Key  |
|---------|--------------------------------------------------|--------------|------------|
| Phase 1 | Planning & Requirement Analysis                  | Week 1 - 3   | Planning   |
| Phase 2 | UI/UX Design                                     | Week 2 - 4   | Design     |
| Phase 3 | Core Backend Development                         | Week 3 - 10  | Backend    |
| Phase 4 | Core Frontend Development                        | Week 5 - 9   | Frontend   |
| Phase 5 | AI Feature Integration                           | Week 6 - 11  | AI         |
| Phase 6 | Real-Time Features & Notifications               | Week 9 - 12  | Real-Time  |
| Phase 7 | Continuous Module-Level Testing                  | Week 5 - 13  | Testing    |
| Phase 8 | Final Integration, Security & Regression Testing | Week 11 - 13 | Testing    |
| Phase 9 | Deployment & Final Presentation                  | Week 9 - 14  | Deployment |

Figure 1: Phase-to-color legend used across the Gantt chart, matching the nine phases in Team Task Timeline & Milestones.

### Gantt Chart — Week-wise Task View

**Dev Barot (202512069) — Project Manager & System Architect**

[illegible]

**Dhrumil Doshi (202512088) — Authentication & User Management Lead**

[illegible]

**Nishka Shah (202512035) — Marketplace & Listing Management Lead**

[illegible]

**Heer Mehta (202512095) — AI & Intelligent Services Lead**

[illegible]



## Heer Shah (202512024) — Search, Chat & Notification Lead

[illegible]

**Charmi Bhayani (202512028) — Rental & Transaction Management Lead**

[illegible]

## Kajal Varlani (202512017) — Admin Panel & Analytics Lead

[illegible]

**Deep Boghara (202512076) — Database, DevOps & Quality Assurance Lead**

[illegible]

## Task Board — Trello

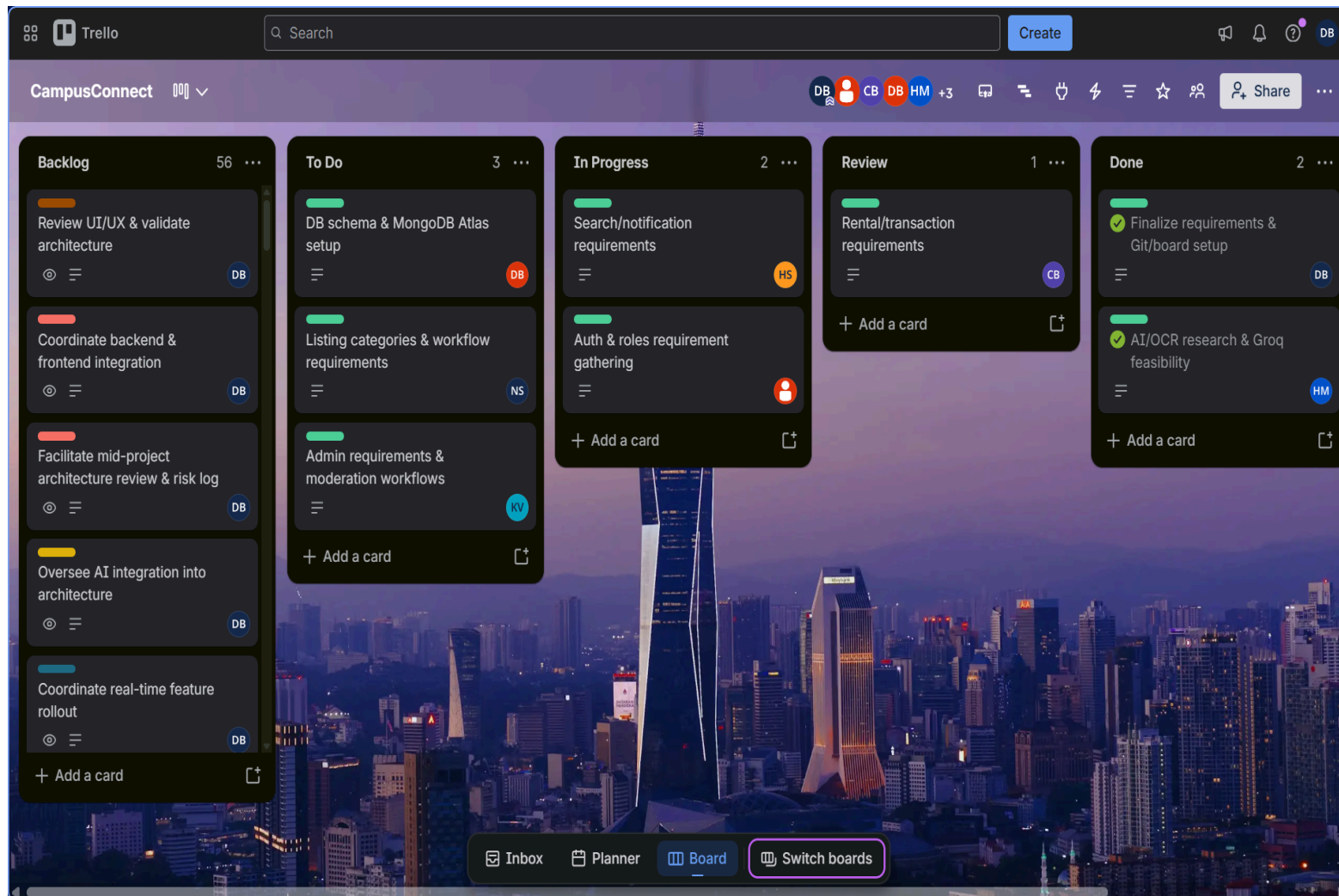


Figure 3: Current state of the CampusConnect Trello board. Backlog holds tasks not yet started; To Do, In Progress, and Review reflect work now underway following the completed planning phase; Done holds the Week 1-2 requirement-gathering tasks completed so far.